

$$\text{change in GPE} = \text{mass} \times \text{change in potential} = 1500(-20+30) \times 10^3 = 15 \text{ GJ}$$

12 C From graph, time to melt completely = $11 \times 60 \text{ s}$

$$\text{Power} \times \text{time} = mL$$

$$0.25 \times 10^3 \times 11 \times 60 = m \times 3.3 \times 10^5$$

$$m = 0.5 \text{ kg}$$

13 A A is correct. Since $\Delta U = Q + W \Rightarrow \Delta U$ can be increased by Q and/or W.